Kipimo

Drones and AI agents that get flood-hit farmers paid the same day

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Kipimo is an embodied agent system for parametric crop insurance after floods. When a river forecast crosses its danger level, a drone surveys the fields where a new photograph could change a payout, three language-model agents coordinate the flight, the farmers' text messages and the review of each claim, a deterministic rule table sets every amount, and an actuary approves each payment before it is released. In a seeded flood on real Tana River terrain in Kenya it paid all ten farmers who were owed, with no wrong amounts.

1 Users and problem

Small farmers lose their crops to floods and often wait weeks for an assessor to visit before their insurance pays out.

- **Users:** smallholder farmers in low-connectivity regions of Africa, Latin America and Asia, the cooperatives and micro-insurers that cover them, and the insurer's actuary who approves each payout.
- **Scale:** there are more than 600 million farms worldwide and most cover under 2 hectares (Lowder et al. 2021); in sub-Saharan Africa fewer than 3 in 100 smallholders hold crop insurance (ISF Advisors 2018).
- **Mechanism:** field-level measurement is slow and costly, so index insurance pays on coarse proxies with basis risk, and late payouts push families to sell assets (Janzen and Carter 2019).

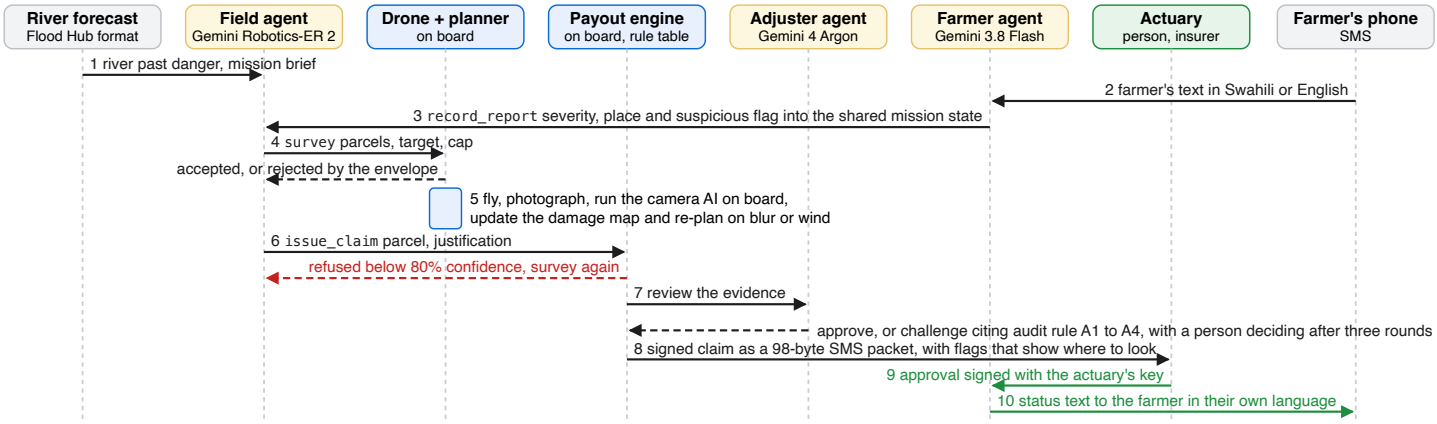
2 Key capabilities

The drone chooses where to look from what its camera has already seen, and it changes its plan whenever conditions change.

- **Event trigger:** a river forecast passing its danger level books the flight, and a wind ensemble picks a safe launch window.
- **Value-weighted sensing:** lanes are chosen greedily by expected reduction in payout uncertainty over a Gaussian-process damage map.
- **Mid-flight adaptation:** a blurred camera brings a lower, slower re-plan, a river breach sends it to the newly flooded fields, wind brings a hold, and the geofence, no-fly zones and battery reserve are hard limits.
- **Low-connectivity delivery:** each claim travels as a signed 98-byte packet that fits a single SMS and waits in a durable queue whenever the link drops.

3 Architecture and agent protocol

The three AI agents send each other messages through a shared mission state and a small set of tools, while plain code flies the drone, sets the money and keeps a signed record of every step.



Solid arrows are requests and dashed arrows are replies. Amber lanes are language-model agents, and each hands its role to an on-board rule policy when the model times out or the link is lost; blue lanes run on the drone or village hub. Every message is an event on one hash-chained, append-only log that the dashboard and the audit both read.

4 Where each model runs, on what hardware, and how fast

Everything that moves the drone or touches money runs on the drone or its village hub, so a mission carries on when the network goes down.

COMPONENT	RUNS ON	MEASURED SPEED AND BUDGET
Camera AI: tiny UNet, int8 ONNX, 0.65 MB	on board Apple M4 CPU on one thread, with NVIDIA Jetson as the target	**1.5 ms** per 64×64 tile in isolation, and a 5 to 12 ms median (21 to 34 ms at p95) in the live run while the 3D view renders.
Damage map: Gaussian process with priors fitted on 30 calibration floods	on board CPU	Updated after every photograph, with 94% of true parcel losses inside the 95% interval after calibration.
Planner and safety envelope: greedy information gain with a (1 − 1/e) guarantee (Krause et al. 2008)	on board CPU	**7 ms** median per plan, and a complete 26-minute mission computes in 0.8 seconds.
Payout engine: YAML rule table, 80% confidence gate, audit rules and Ed25519 signing	on board CPU	Deterministic and under 1 ms per claim, so identical evidence always yields the identical amount.
Agents: Robotics-ER 2 (Field), Gemini 4 Argon (Adjuster), Gemini 3.8 Flash (Farmer)	cloud API whenever a link is available	2.5 to 11 s per turn within 45 tool calls, 300 s per role and 20 min per mission, after which each role continues as an on-board rule policy; a live mission cost US$0.36.
River and wind forecasts in the formats of Google Flood Hub and WeatherNext	cloud before take-off	Hourly series read before launch and cached on the hub, simulated in this prototype.

5 Results

Across seeded floods the system pays the right farmers the right amounts while flying about a third less than a sweep of every field.

- **Live Gemini mission (seed 7):** all 10 owed farmers were paid with zero wrong amounts, the first claim came 4 minutes after the danger line, and two Argon challenges cleared after a re-flight.
- **Unscripted river breach mid-mission:** the drone turned at once, and all 13 farmers then owed were paid correctly.
- **30 seeded floods:** the planner flew a median of 10.2 km with a mean payout error of 0.85%, and a full sweep flew 15.9 km with 0.89%, so the planner flew 36% less.
- **60 evaluation floods with on-board roles:** zero unreviewed claims, zero flights above the wind limit or into no-fly zones, and every injected instruction flagged.

BUILT WITH

Gemini 4 Argon · Gemini Robotics-ER 2 · Gemini 3.8 Flash
Google Groundsource · NVIDIA Jetson · Google DeepMind WeatherNext format
Google Flood Hub format · Copernicus GLO-30 · Sentinel-2 (EOX)
ONNX Runtime · PyTorch · Ed25519 signing

6 Critical failure mode and safety

The most serious failure would be a wrong payment, whether too large, too small, sent to the wrong farmer or based on weak evidence, so six independent checks sit between a model's request and any money moving.

- Language models act through typed tools, and deterministic code executes every flight command and computes every amount.
- The payout engine refuses any claim below 80% confidence, and the Field agent must then send the drone back for more evidence.
- The Adjuster can challenge a claim by citing a written audit rule that the evidence satisfies, and after three rounds the case passes to a person.
- The actuary signs every approval with a personal key, guided by flags such as payouts above half the sum insured.
- Farmers' messages feed into where the drone looks next, injected instructions are flagged, and every claim and event is signed and hash-chained so the audit detects any altered number.
- When conditions are uncertain the system falls back safely, with blur leading to a lower re-plan, wind to a hold, a lost link to a queue and an unreachable model to its rule policy.

7 Limits

- The terrain (Copernicus GLO-30) and imagery (Sentinel-2) are real, while the flood, the crops, the drone and the payments are simulated.
- The camera AI was trained on synthetic tiles and still needs validation on real drone imagery and on Jetson hardware.
- The language-model agents need a network link, and without one the rule policies run with less flexibility on unusual messages.
- The river and wind forecasts are simulated until the Flood Forecasting API and WeatherNext data are connected.

References: Nearing et al., Nature 627, 2024 · Price et al., Nature 637, 2025 · Gemini Robotics Team, arXiv 2503.20020, 2025 · Krause, Singh and Guestrin, JMLR 9, 2008 · Carter et al., Annu. Rev. Resour. Econ. 9, 2017 · Janzen and Carter, AJAE 101, 2019 · Lowder, Sánchez and Bertini, World Development, 2021 · ISF Advisors, 2018.